

HB-Adaptive model: learning diagnostics and data comparison

Remote branch: model-verification

Model: HB-AdaptiveConfidenceObserver (registry key: hb_adaptive)

Core parameters fitted per subject: $k_{\text{like}}[0.06]$, $k_{\text{like}}[0.12]$, $k_{\text{like}}[0.24]$, k_{motor} , p_{random} , lam .

Latent learning replayed from feedback: joint belief $b_t(\kappa, \alpha)$.

Subjects plotted: 12; converged fits: 12/12.

Block alignment check: 0 session/run blocks contain more than one prior_std.

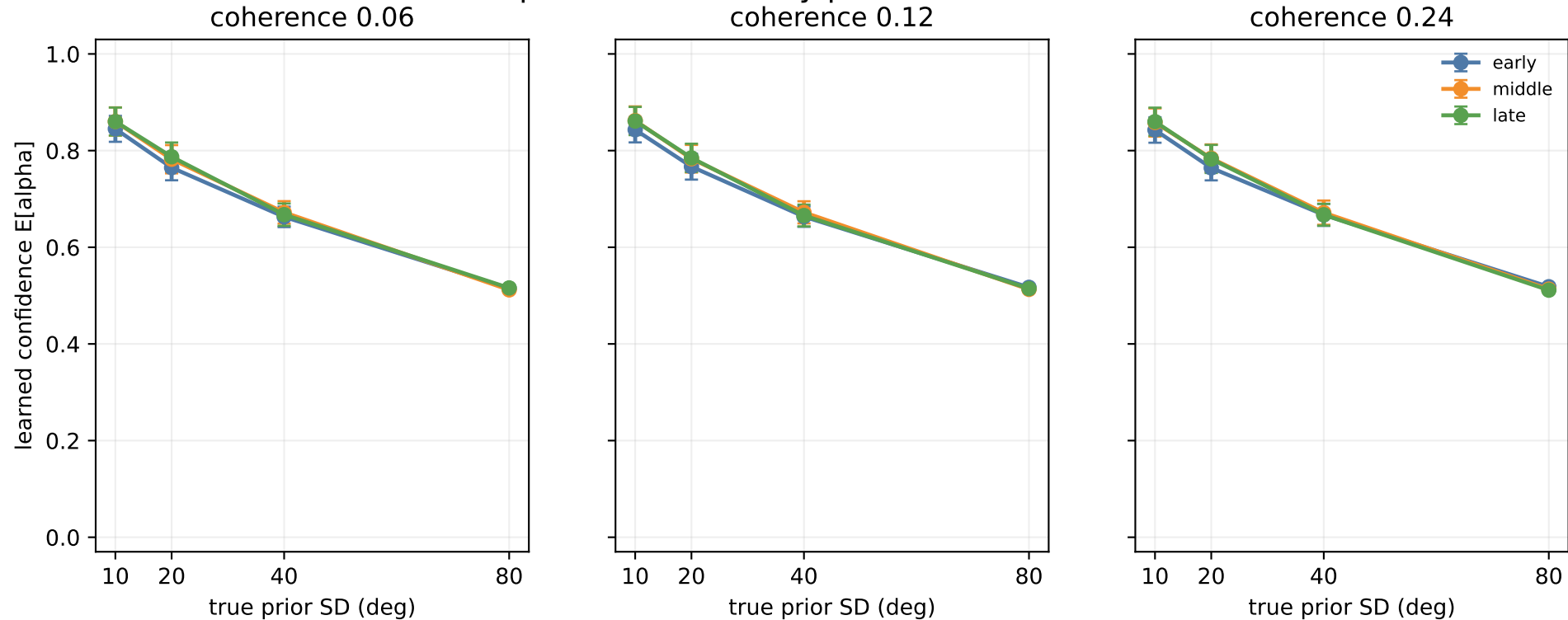
Important interpretation note:

The hidden belief update uses feedback directions, not coherence. Coherence changes the sensory likelihood and prediction. Therefore coherence-stratified learning plots show where trials of each coherence fall on the same learned trajectory.

ANOVA-style fixed-subject tests are computed on subject means by coherence x prior_std.

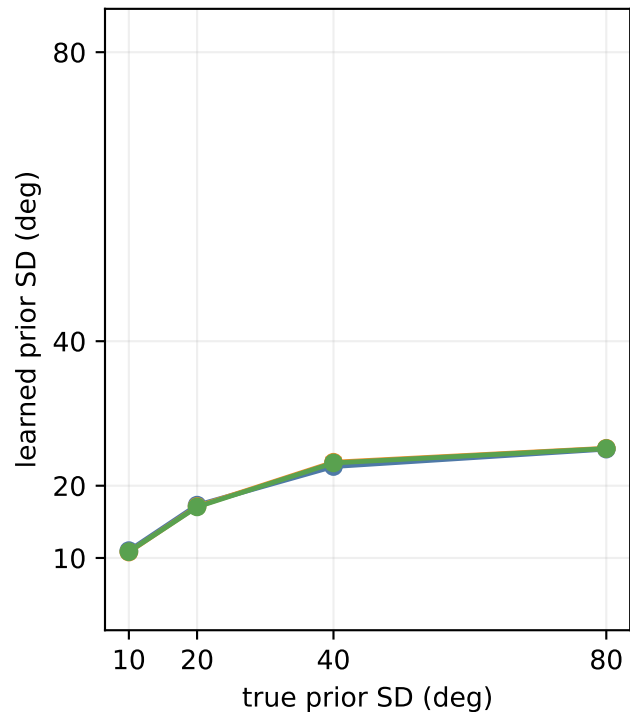
measure	effect	df_effect	df_error	F	p
mean_believed_alpha	coherence	2	127	0.00	0.998
mean_believed_alpha	prior_std	3	127	489.97	0
mean_believed_alpha	coherence:prior_std	6	121	0.00	1
mean_believed_sd	coherence	2	127	0.01	0.991
mean_believed_sd	prior_std	3	127	1413.72	0
mean_believed_sd	coherence:prior_std	6	121	0.03	1

Learned prior confidence by prior width and coherence

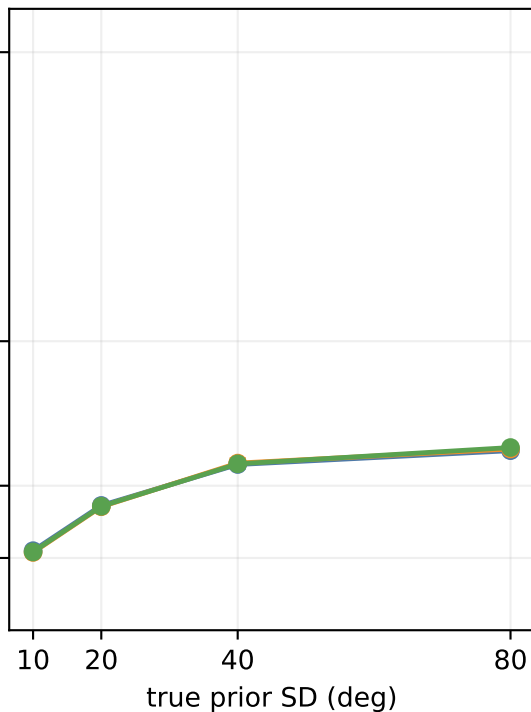


Learned prior width by prior width and coherence

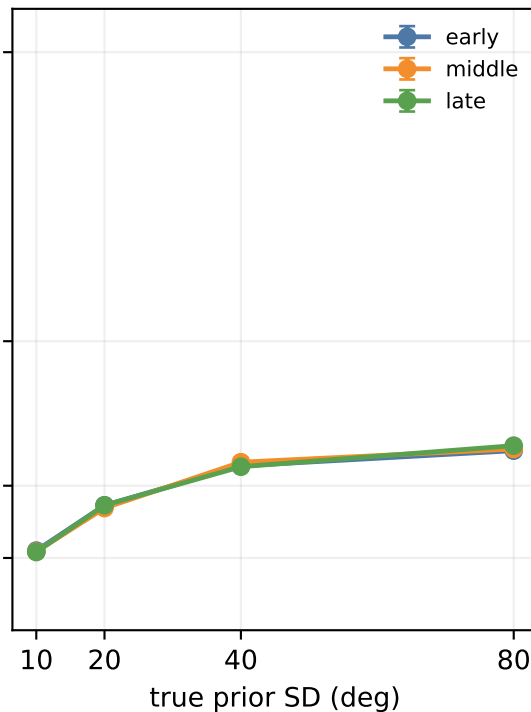
coherence 0.06



coherence 0.12

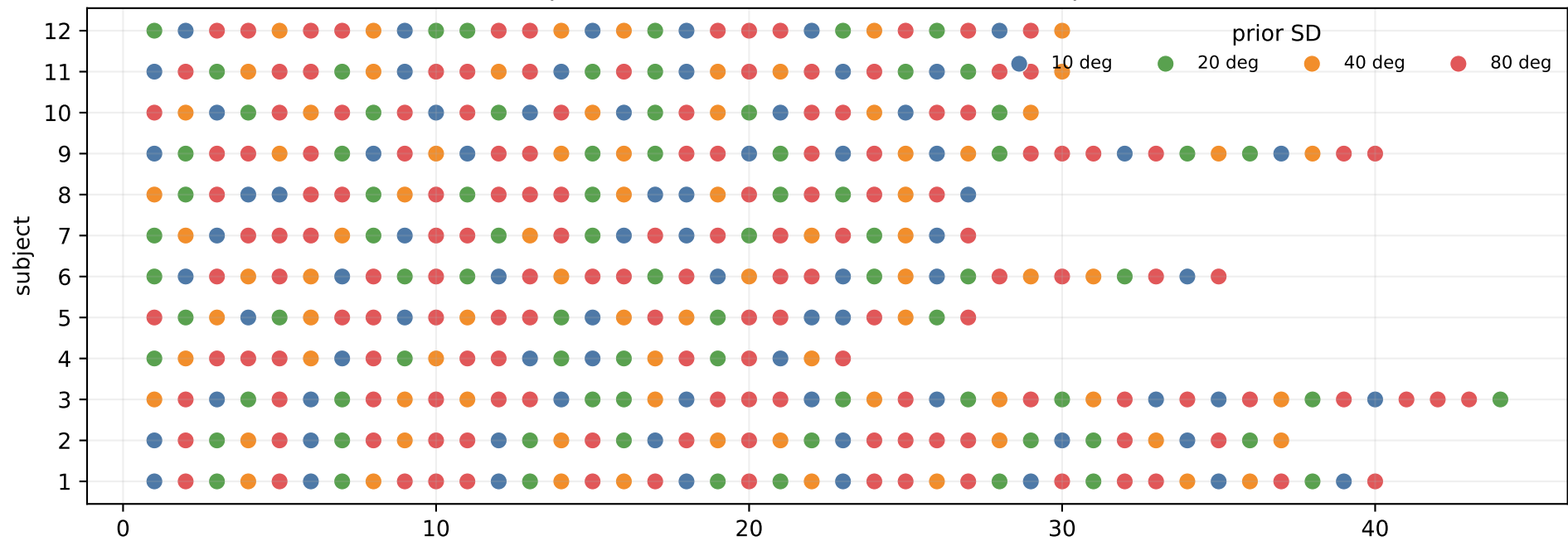


coherence 0.24

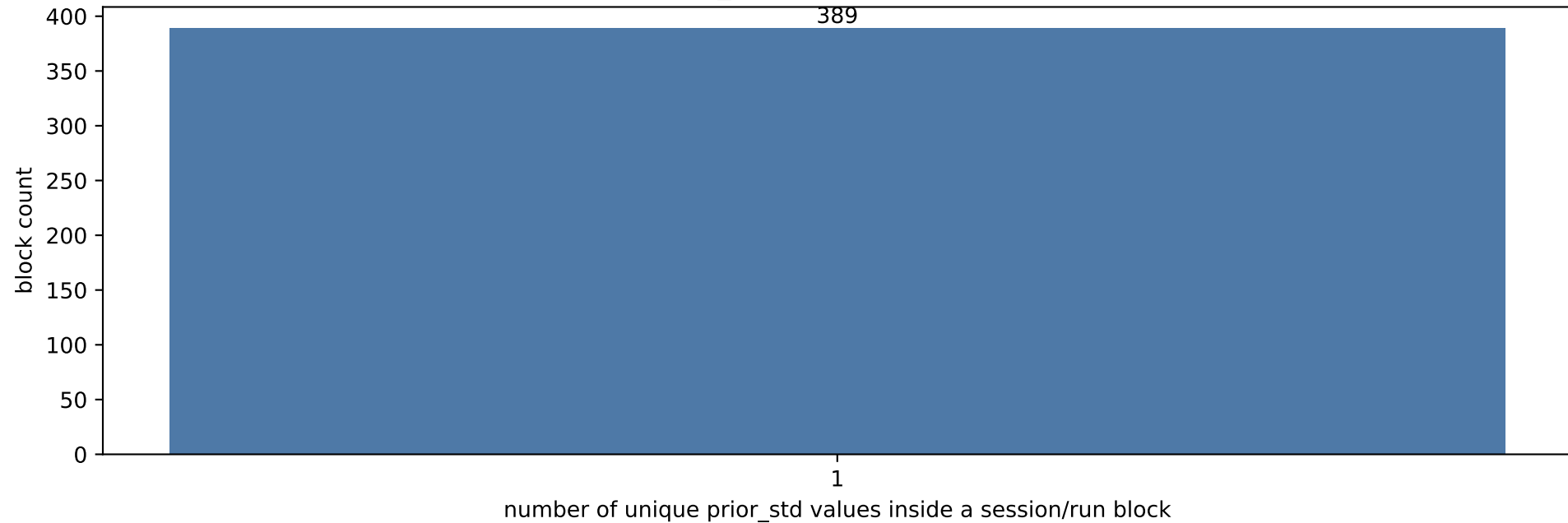


Block alignment diagnostic

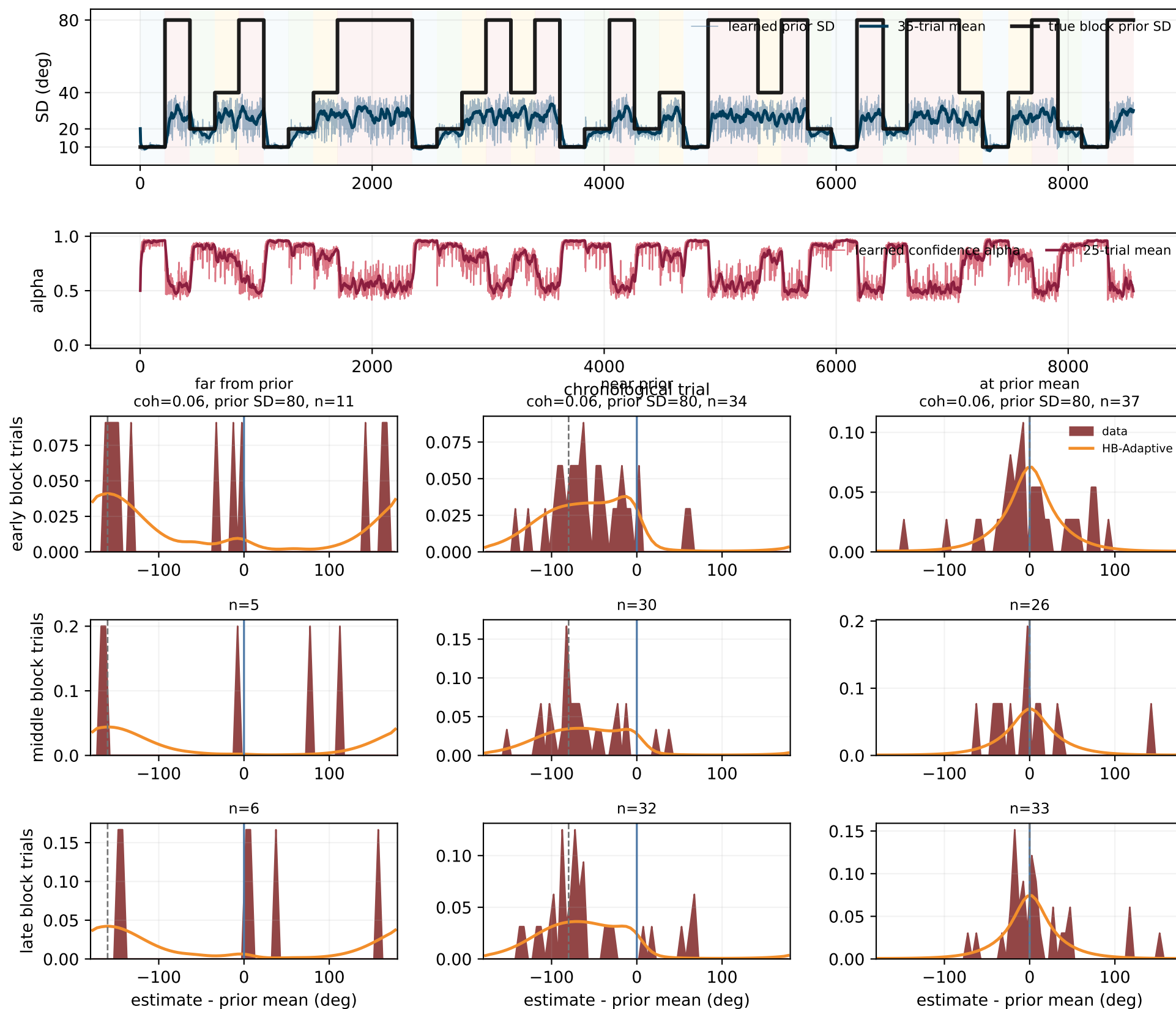
Each point is one run block; color is the true prior SD



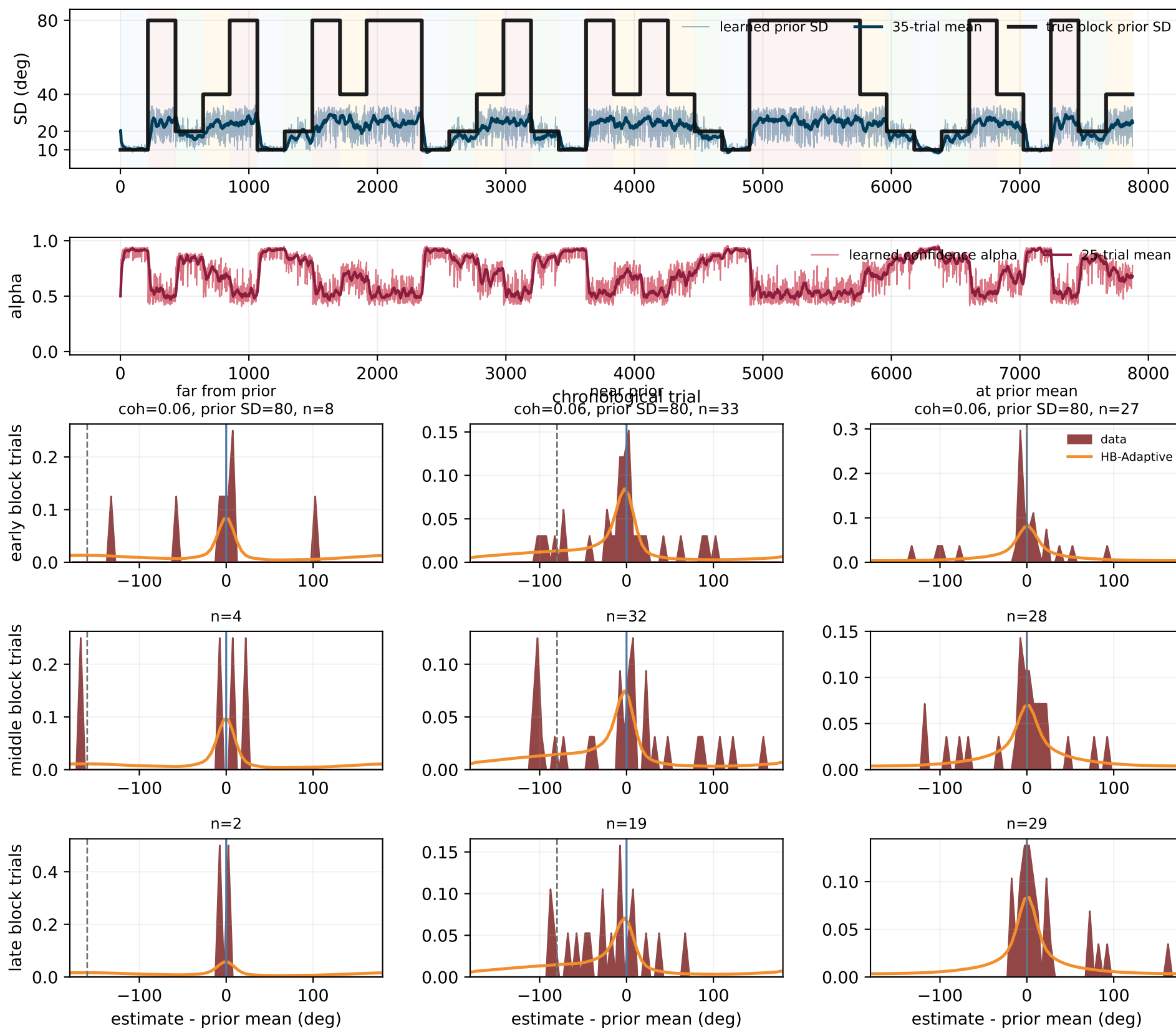
If any block has >1 unique prior_std, block definitions may be shifted or too broad



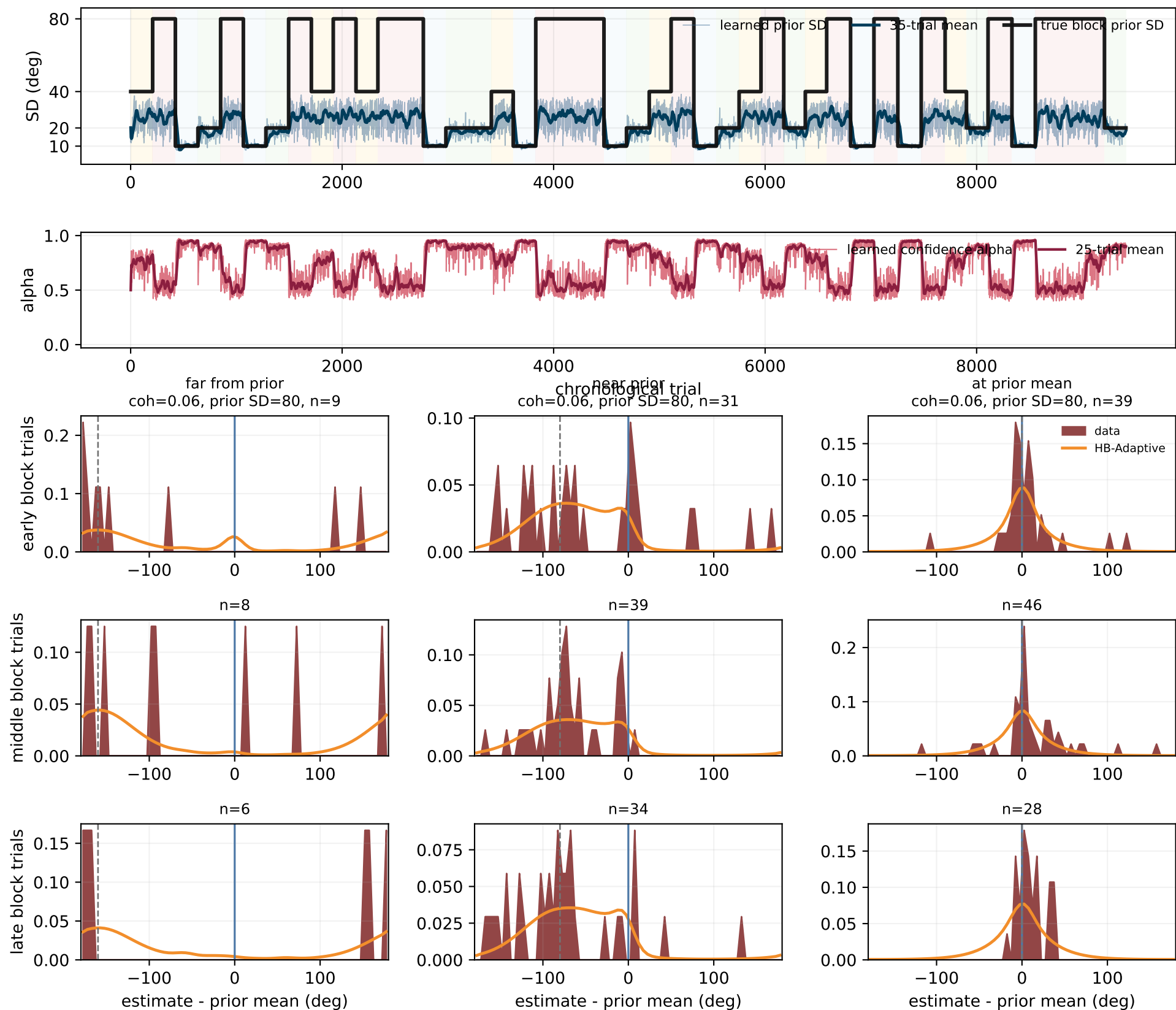
Subject 1: HB-Adaptive predictions vs responses by block phase



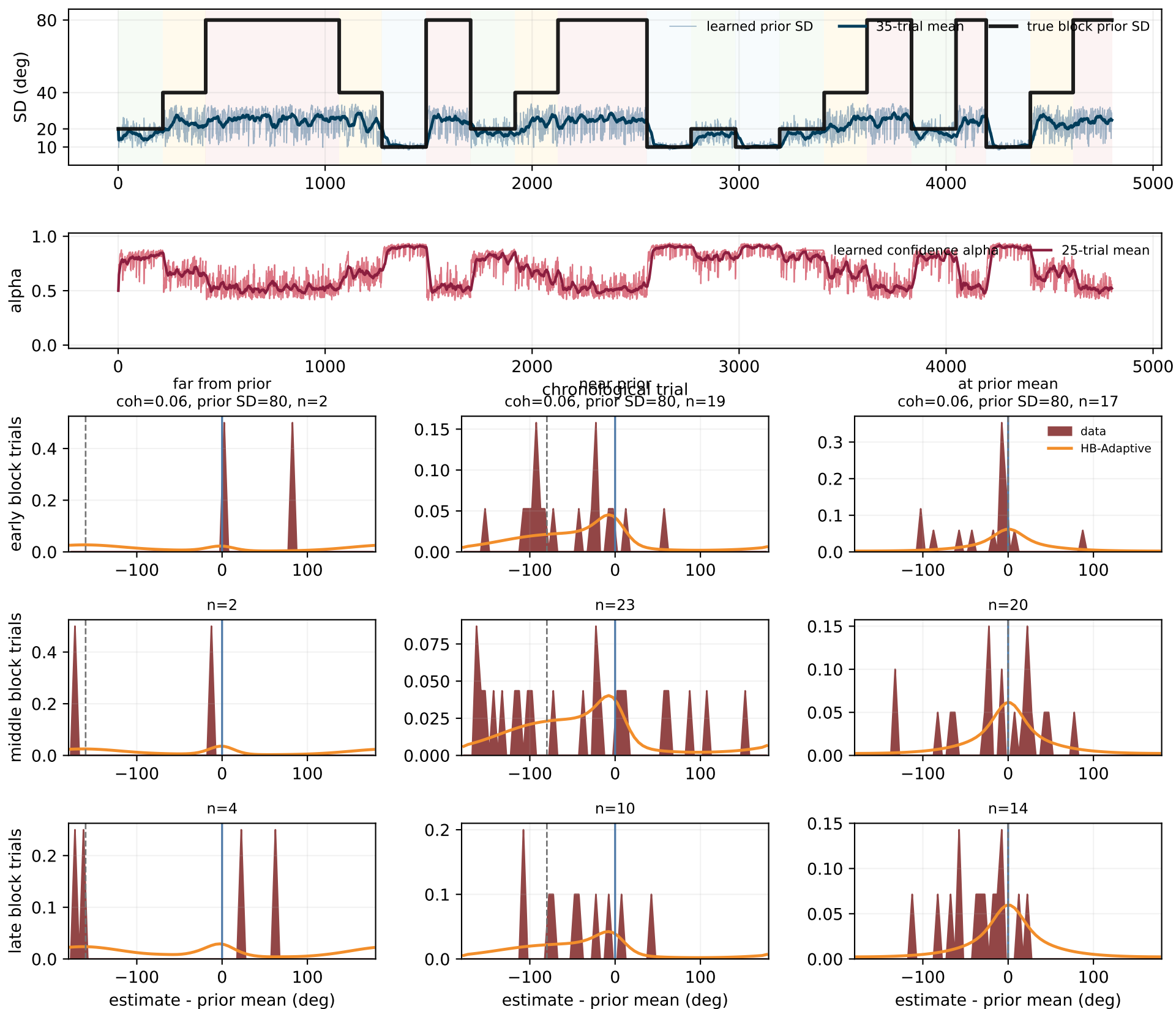
Subject 2: HB-Adaptive predictions vs responses by block phase



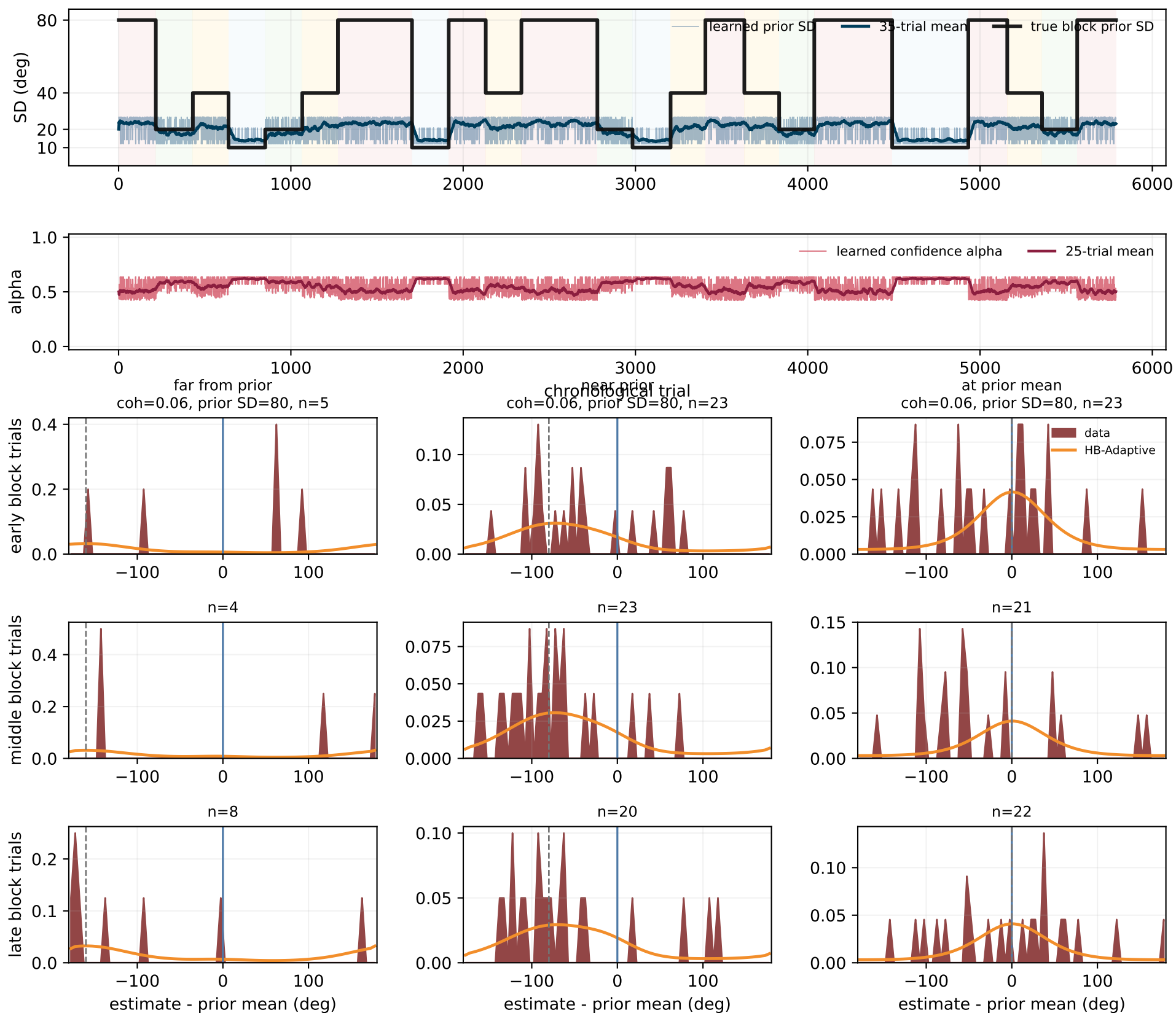
Subject 3: HB-Adaptive predictions vs responses by block phase



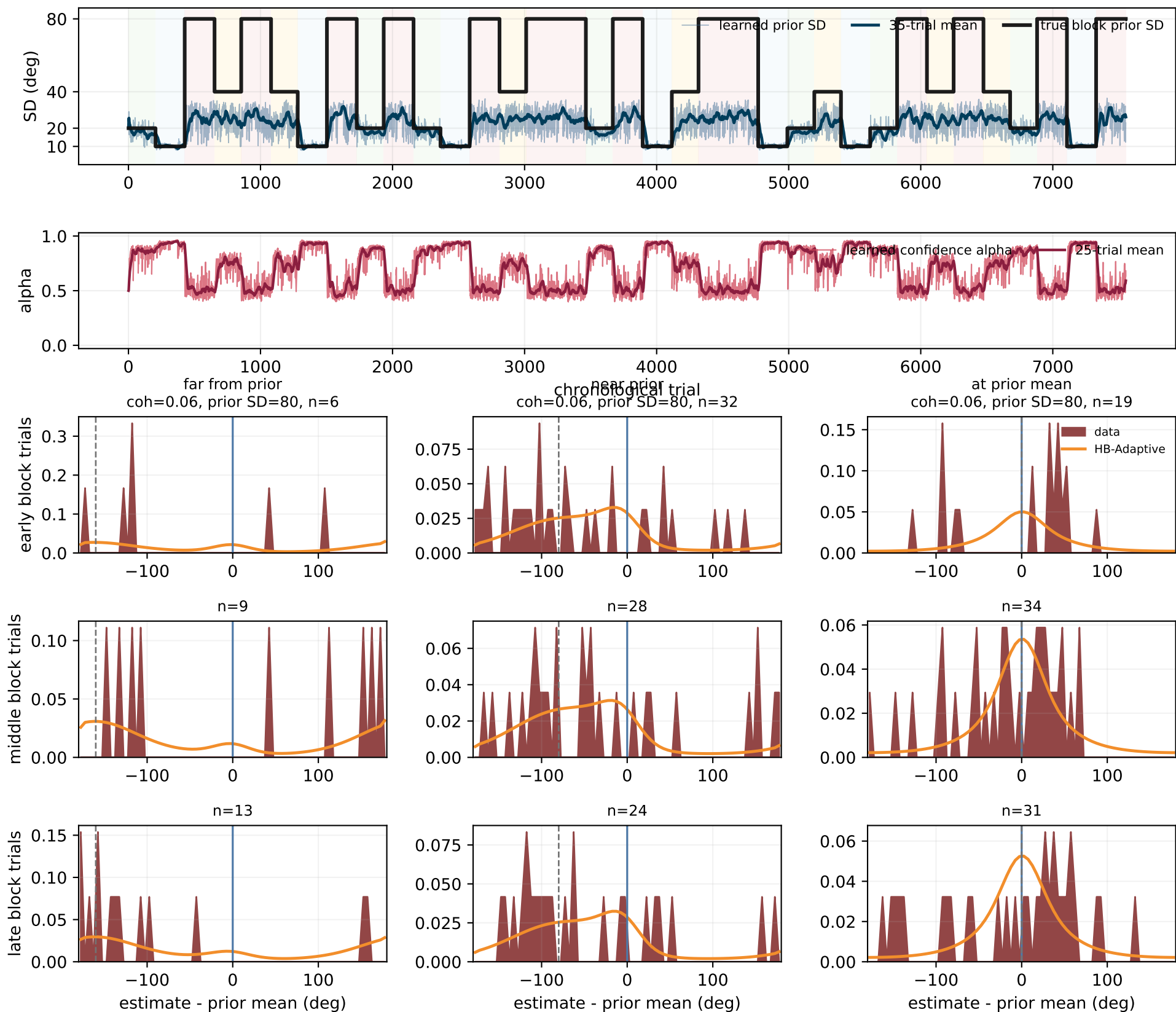
Subject 4: HB-Adaptive predictions vs responses by block phase



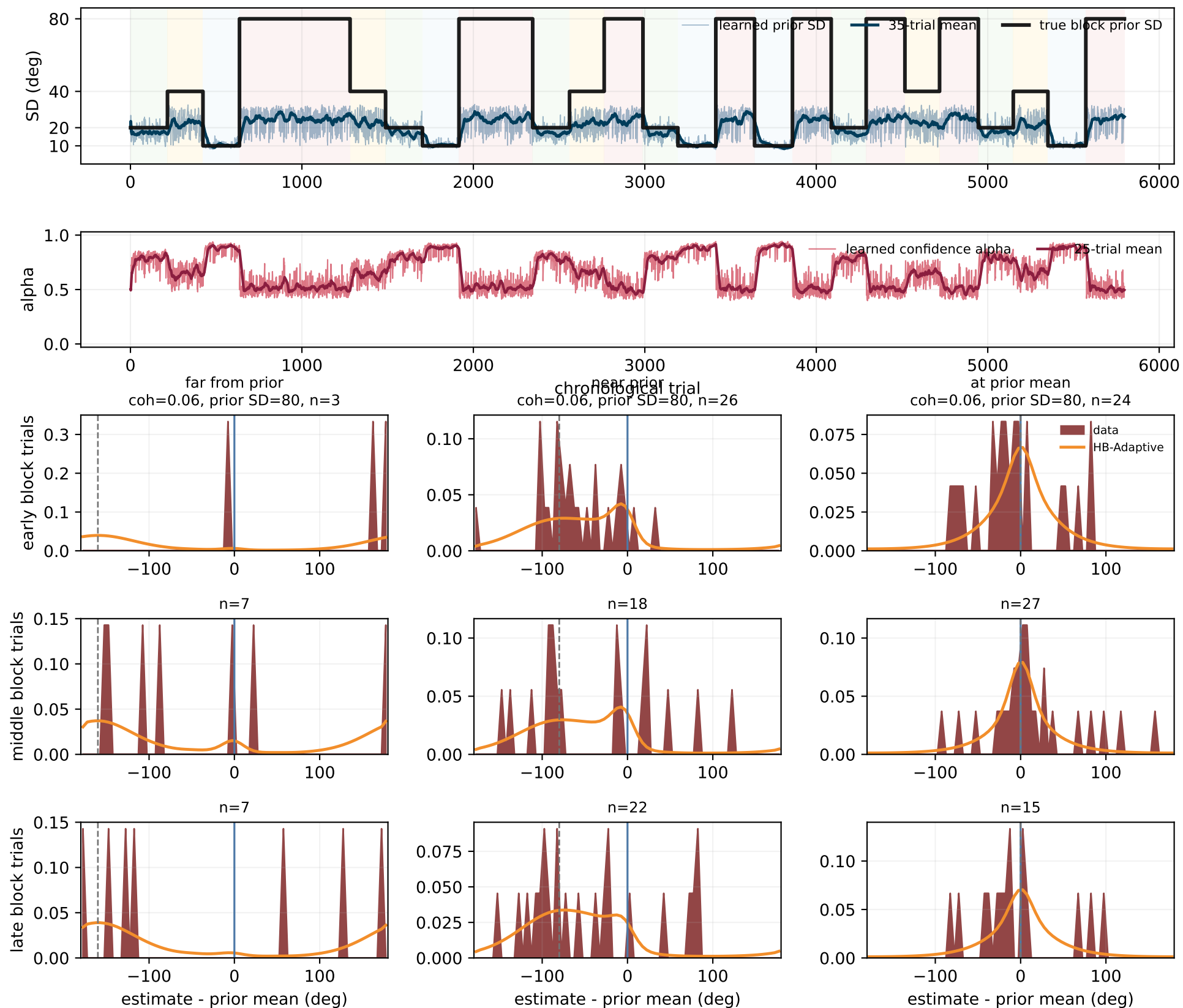
Subject 5: HB-Adaptive predictions vs responses by block phase



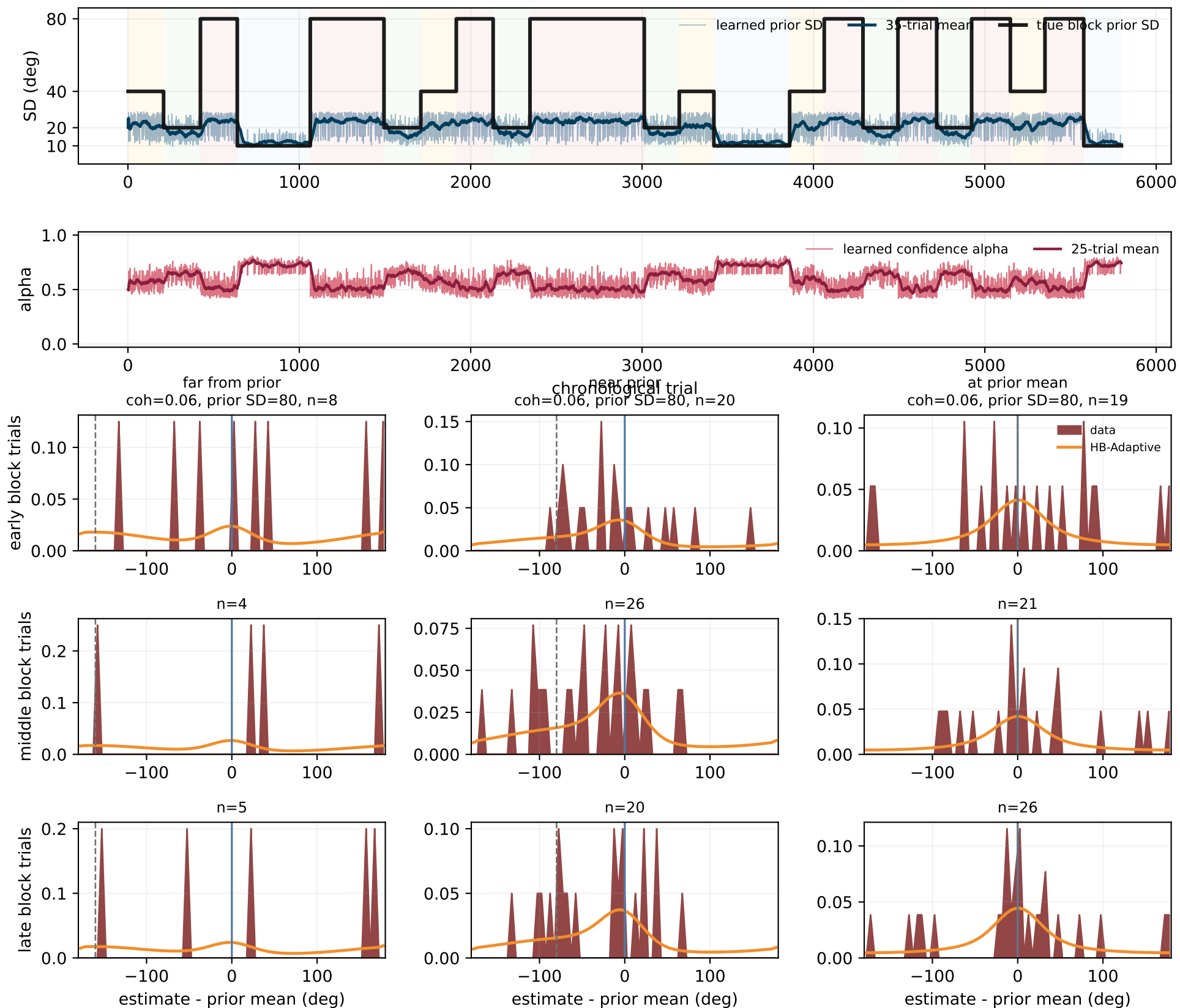
Subject 6: HB-Adaptive predictions vs responses by block phase



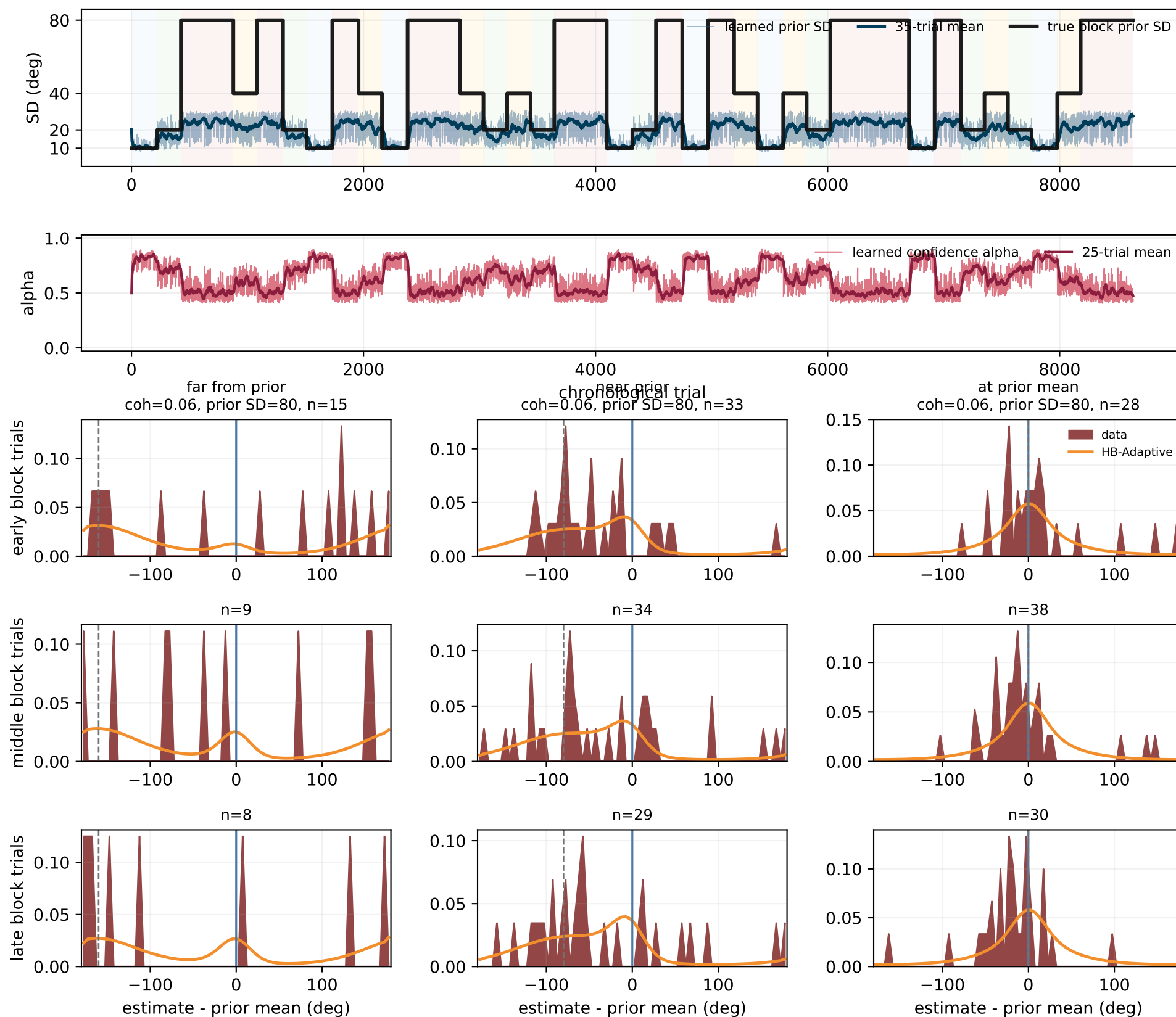
Subject 7: HB-Adaptive predictions vs responses by block phase



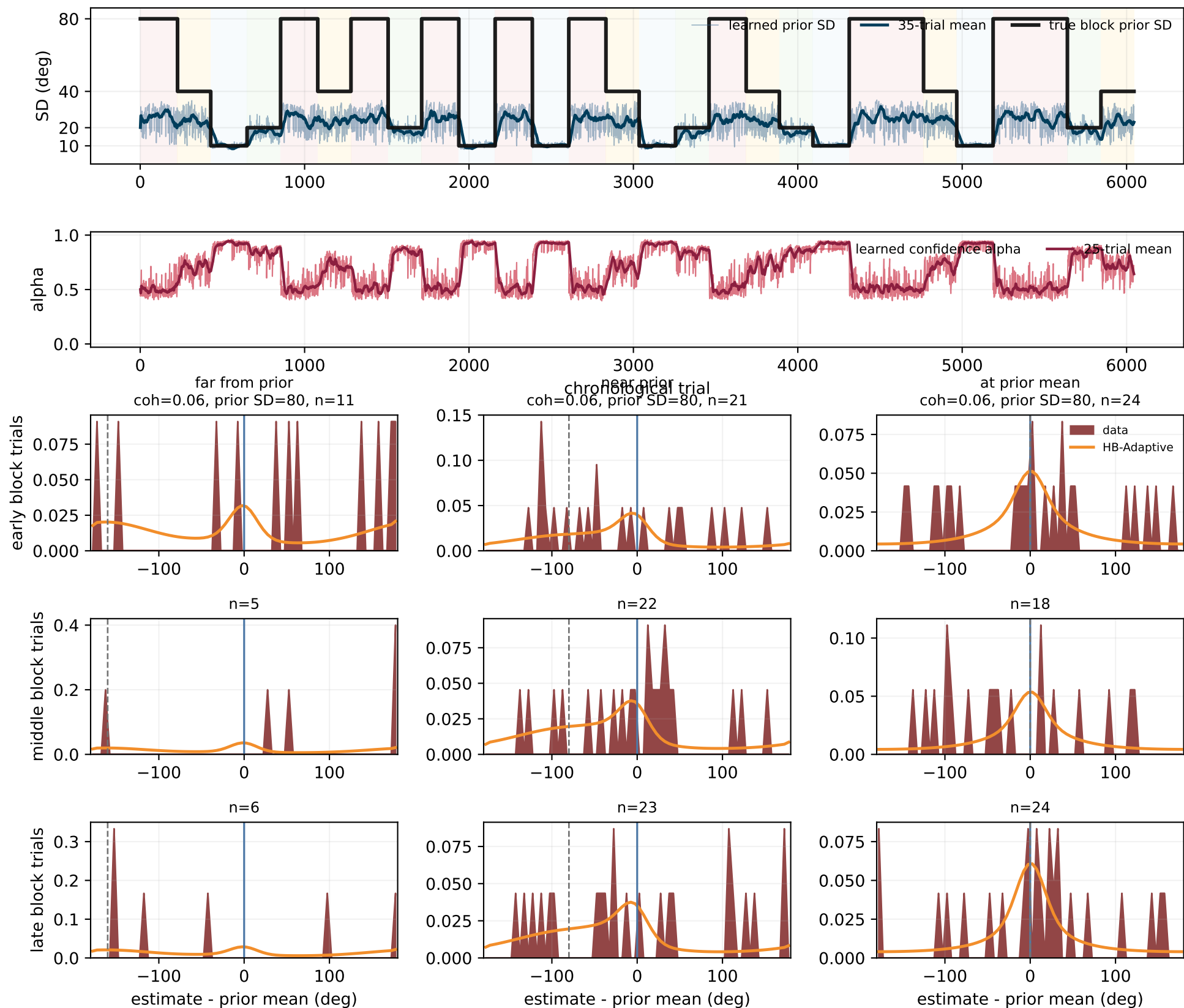
Subject 8: HB-Adaptive predictions vs responses by block phase



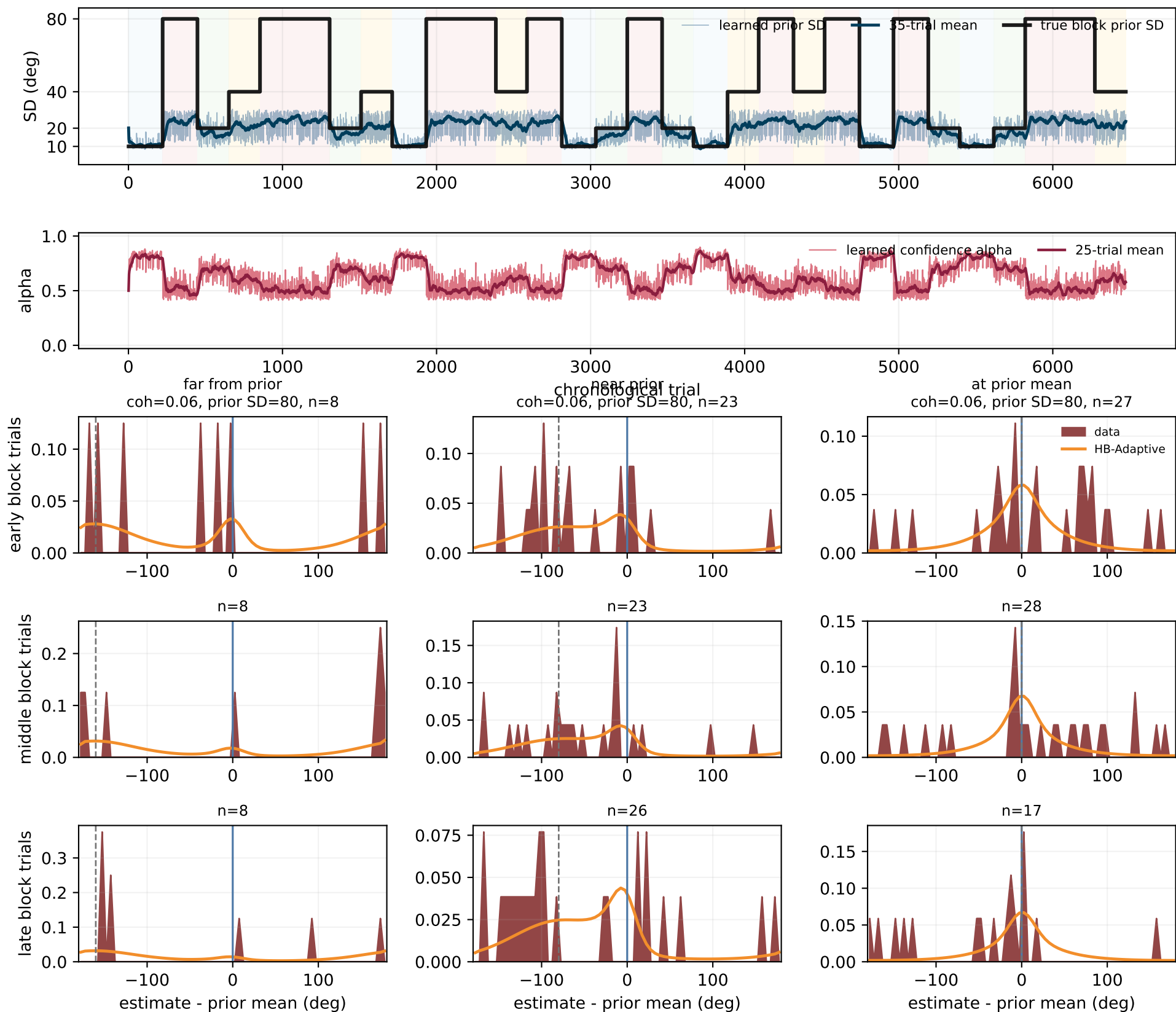
Subject 9: HB-Adaptive predictions vs responses by block phase



Subject 10: HB-Adaptive predictions vs responses by block phase



Subject 11: HB-Adaptive predictions vs responses by block phase



Subject 12: HB-Adaptive predictions vs responses by block phase

